

Set 7: Pattern Printing

Questions & Answers

Question 1

Problem: Print the following pattern:

```
#  
##  
###  
####  
#####
```

Answer:

```
for i in range(1, 6):  
    print("#" * i)
```

Output:

```
#  
##  
###  
####  
#####
```

Question 2

Problem: Print the following pattern:

```
#####  
####  
###  
##  
#
```

Answer:

```
for i in range(5, 0, -1):  
    print("#" * i)
```

Output:

```
#####  
####  
###  
##  
#
```

Question 3

Problem: Print the following pattern:

```
*
**
***
****
*****
****
***
**
*
```

Answer:

```
for i in range(1, 6):
    print("*" * i)

for i in range(4, 0, -1):
    print("*" * i)
```

Output:

```
*
**
***
****
*****
****
***
**
*
```

Question 4

Problem: Print the following pattern:

```
1
12
123
1234
12345
```

Answer:

```
for i in range(1, 6):
    for j in range(1, i + 1):
        print(j, end="")
    print()
```

Output:

```
1
12
123
1234
12345
```

Question 5

Problem: Print the following pattern:

```
12345
1234
123
12
1
```

Answer:

```
for i in range(5, 0, -1):
    for j in range(1, i + 1):
        print(j, end="")
    print()
```

Output:

```
12345
1234
123
12
1
```

Question 6

Problem: Print the following pattern:

```
A
AB
ABC
ABCD
ABCDE
```

Answer:

```
for i in range(1, 6):
    for j in range(i):
        print(chr(65 + j), end="")
    print()
```

Output:

```
A
AB
ABC
ABCD
ABCDE
```

Notes:

```
Concept: chr()
chr(65) # A
chr(66) # B
chr(67) # C
```

Question 7

Problem: Print the following pattern:

```
1
22
333
4444
55555
```

Answer:

```
for i in range(1, 6):
    print(str(i) * i)
```

Output:

```
1
22
333
4444
55555
```

Question 8

Problem: Print the following pattern:

```
54321
5432
543
54
5
```

Answer:

```
for i in range(5, 0, -1):
    for j in range(5, 6 - i, -1):
        print(j, end="")
    print()
```

Output:

```
5432
543
54
5
```

Challenge Question

Problem: Print the following pattern:

```
#
##
###
####
#####
```

Answer:

```
for i in range(1, 6):
    print(" " * (5 - i) + "#" * i)
```

Output:

```
#
##
###
####
#####
```

Notes:

Logic:
Spaces = 5 - i
Hashes = i

Example, when i = 2:
Spaces = 5 - 2 = 3, Hashes = 2
So:
##

Bonus Challenge

Problem: Print the following pattern:

```
1
12
123
1234
12345
```

Answer:

```
for i in range(1, 6):
    print(" " * (5 - i), end="")

    for j in range(1, i + 1):
        print(j, end="")

    print()
```

Output:

```
1
12
123
1234
12345
```